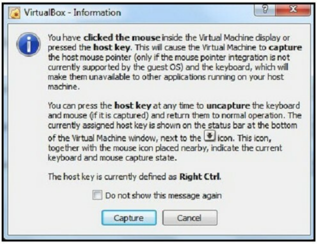

Figure 9: VirtualBox mouse capture notification

Use-case 6: Create a virtual lab to learn enterprise software

This is by far one of my favourite use cases of VirtualBox. Being a curious Tom, I love to get my hands dirty by learning the enterprise software used at work—mail server software, directory software, whatever. VirtualBox is a wonderful resource in such cases. You can use it to create a mini-virtual lab consisting of server(s) and a few client machines. You can then deploy the software you wish to learn in this virtual lab and experiment with it.

Best practices and recommendations

A few tips and tricks can go a long way to help you get the best out of VirtualBox. Based on my experience with it, I am listing a few below.

Use a 64-bit processor: Usually, a machine with a 32-bit processor limits the usable RAM to 3 GB. This, in turn, limits the amount of RAM you can assign to a VM and also the number of VMs you can create. On the other hand, a 64-bit processor gives you the flexibility to use up to 16 GB of RAM on the host system, thereby increasing the amount of RAM that can be allocated to VMs.

Enable virtualisation in the BIOS: As stated earlier, most of the systems have this feature disabled, by default. If you wish to create 64-bit VMs, you'll need to enable this option in the BIOS.

Use NAT: Unless you're planning to create an internal network, use NAT as a network mode. This would save you the trouble of configuring the VM to access the Internet, because with NAT VMs hide behind the IP address of the host machine and can access the Internet as long as host system has access to it.

Use auto-mount for shared folders: Keeping the auto-mount setting checked while creating the shared folder will ensure that the folder is visible as soon as the VM boots up.

Use a separate partition or hard drive: By default, VirtualBox stores the VM files in the C drive. This can fill the drive pretty fast, if you're working with multiple VMs. In order to keep the C drive clean, it is recommended that you use either a separate partition or a hard drive to store the VMs.

Install guest additions: VirtualBox guest additions install certain drivers on the VM, which enable it to function like a normal machine. For example, without guest additions installed, a VM will not occupy the full screen even in the *Full Screen* mode.

RAM allocation: Never allocate more than 75 per cent of available RAM to a single VM or else the host system's performance will be affected. You must keep RAM allocation within the green region in the *Memory Size* window.

Common errors and resolutions

Error: VT-x/AMD-V hardware acceleration has been enabled, but is not operational

This error comes up when you're trying to boot a 64-bit VM but VirtualBox isn't able to find hardware virtualisation capabilities on the host machine. This can be fixed by turning on the VT - x or *Virtualization* option in the BIOS.

Error: 0x0000225

This error is related to I/O APIC (Advanced Programmable Interrupt Controllers) settings. APIC provides a mechanism by which the devices attached to the VM communicate with the processor. If you're using a 64-bit VM, I/O APIC settings need to be enabled. You can enable this by going to *Settings > System* and checking 'Enable I/O APIC' in the *Extended Features* section.

Error: Error loading operating system

This is the most common type of error that users face when migrating Windows VMs from VMware to VirtualBox. This can be fixed easily by changing the controller of the virtual hard drive from SCSI to IDE and then repairing the boot manager using the installation disk. The link *http://goo.gl/PzOAyo* contains a detailed tutorial on fixing this error.

Besides the above mentioned use cases, virtualisation technology also makes it possible to optimally use hardware. Organisations are today implementing a virtualised infrastructure for their employees to work in. This not only reduces the amount of e-waste but also serves as a secure computing environment. Even home users can efficiently utilise their old hardware with the help of virtualisation tools like VirtualBox.

If you have any other use case for VirtualBox, or if you use VirtualBox to tackle any other day-to-day challenges, do share them. END

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